

Differentiable functions – 5

Convexity as a method: Jensen, Hermite–Hadamard,
Karamata

Homework

1 сентября 2026 года

Theoretical minimum. Jensen’s inequality with weights and in integral form; power means and Hölder; Hermite–Hadamard; majorisation and Karamata’s inequality; Popoviciu’s inequality; the tangent line trick; the SOS and smoothing methods; convexity of $x \mapsto x^p$, $-\ln x$, e^x and of the functions appearing in triangle inequalities.

1. Prove Popoviciu’s inequality: for convex f and reals x, y, z , $f(x) + f(y) + f(z) + 3f\left(\frac{x+y+z}{3}\right) \geq 2\left[f\left(\frac{x+y}{2}\right) + f\left(\frac{y+z}{2}\right) + f\left(\frac{z+x}{2}\right)\right]$.
2. Prove the inequality between the arithmetic, geometric and harmonic means by three different convexity arguments: Jensen, the tangent line trick and smoothing.
3. Let $a, b, c > 0$ with $a + b + c = 1$. Prove that $\frac{a}{1+bc} + \frac{b}{1+ca} + \frac{c}{1+ab} \geq \frac{9}{10}$.
4. Prove that in every triangle $\frac{1}{\sin^2 A} + \frac{1}{\sin^2 B} + \frac{1}{\sin^2 C} \geq 4$, and determine when equality holds.
5. Prove Chebyshev’s sum inequality from majorisation, and deduce that for a convex f and $a_1 \leq \dots \leq a_n$ the sequence $(f(a_i))$ is majorised by $(f(a_1), \dots, f(a_n))$ rearranged in a suitable way.
6. Prove that a midpoint convex function bounded above on a set of positive measure is convex, and give an example of a midpoint convex function which is not convex.
7. Let f be convex on $[0, 1]$ with $f(0) = f(1) = 0$. Prove that $\int_0^1 |f| \leq \frac{\max |f|}{2}$ and find the extremal functions.

8. Prove the integral Jensen inequality $\varphi(\int_0^1 g) \leq \int_0^1 \varphi \circ g$ for convex φ and integrable g , and deduce the inequality between $\int e^g$ and $e^{\int g}$.
9. Let f be convex on $\mathbb{R}$ and bounded above. Prove that f is constant; give the analogous statement for concave functions bounded below.
10. Prove that a convex function on $\mathbb{R}^n$ which is bounded on a ball is continuous, and give an example of a discontinuous convex function on an infinite-dimensional space.